

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
```

```
df = pd.read_csv('/content/zomato.csv')
df.head()
```

	address	name	online_order	book_table	rate	votes	phone	location	re
0	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775	080 42297555\r\n+91 9743772233	Banashankari	
1	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1/5	787	080 41714161	Banashankari	
2	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8/5	918	+91 9663487993	Banashankari	
3	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7/5	88	+91 9620009302	Banashankari	
4	10, 3rd Floor, Lakshmi Associates, Gandhi Baza...	Grand Village	No	No	3.8/5	166	+91 8026612447\r\n+91 9901210005	Basavanagudi	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.isnull().sum()
```

	0
address	17
name	16
online_order	19
book_table	58
rate	7838
votes	78
phone	1296
location	126
rest_type	338
dish_liked	28225
cuisines	203
approx_cost(for two people)	521
listed_in(type)	4610

dtype: int64

```
df.fillna(method='ffill', inplace=True)
```

```
/tmp/ipykernel_1279/3970806690.py:1: FutureWarning: DataFrame.fillna with 'method' is deprecated and will be removed in a future version.
  df.fillna(method='ffill', inplace=True)
```

```
df.drop_duplicates(inplace=True)
```

```
df['rate'] = df['rate'].astype(str)
df['rate'] = df['rate'].str.replace('/5','')
df['rate'] = pd.to_numeric(df['rate'], errors='coerce')
```

```
print(df.columns)
```

```
Index(['address', 'name', 'online_order', 'book_table', 'rate', 'votes',
      'phone', 'location', 'rest_type', 'dish_liked', 'cuisines',
      'approx_cost(for two people)', 'listed_in(type)'],
      dtype='object')
```

```
df.columns.tolist()
```

```
['address',
 'name',
 'online_order',
 'book_table',
 'rate',
 'votes',
 'phone',
 'location',
 'rest_type',
 'dish_liked',
 'cuisines',
 'approx_cost(for two people)',
 'listed_in(type)']
```

Double-click (or enter) to edit

```
df['approx_cost(for two people)'] = pd.to_numeric(
    df['approx_cost(for two people)'].astype(str).str.replace(',',' '),
    errors='coerce'
)
```

```
print(df.columns.tolist())
```

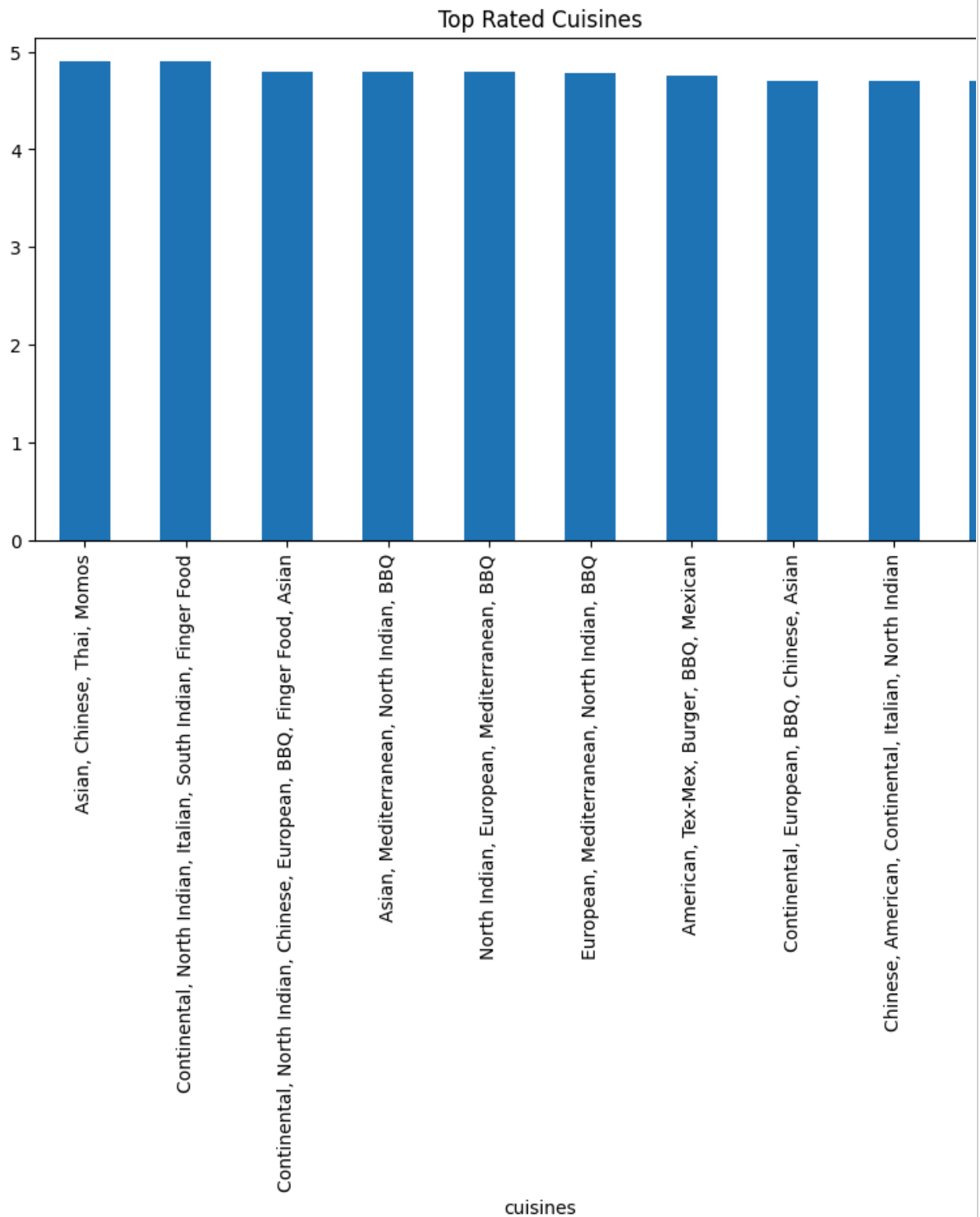
```
['address', 'name', 'online_order', 'book_table', 'rate', 'votes', 'phone', 'location', 'rest_type']
```

```
for col in df.columns:
    if 'cuisine' in col.lower():
        print(col)
```

```
cuisines
```

```
cuisine_rating = df.groupby('cuisines')['rate'].mean().sort_values(ascending=False).head(10)
```

```
plt.figure(figsize=(10,5))
cuisine_rating.plot(kind='bar')
plt.title("Top Rated Cuisines")
plt.show()
```



```
print(df.columns.tolist())
```

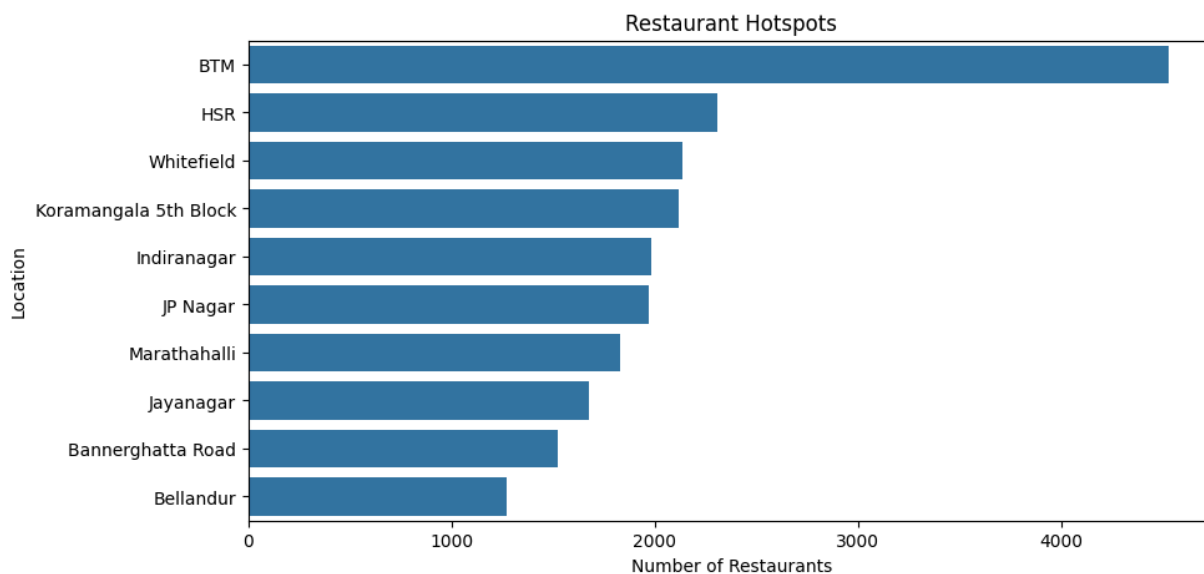
```
ocation', 'rest_type', 'dish_liked', 'cuisines', 'approx_cost(for two people)', 'listed_in(type)']
```

```
for col in df.columns:
    if any(word in col.lower() for word in ['city','location','locality','address','place']):
        print(col)
```

```
address
location
```

```
top_locations = df['location'].value_counts().head(10)
```

```
plt.figure(figsize=(10,5))
sns.barplot(x=top_locations.values, y=top_locations.index)
plt.title("Restaurant Hotspots")
plt.xlabel("Number of Restaurants")
plt.ylabel("Location")
plt.show()
```

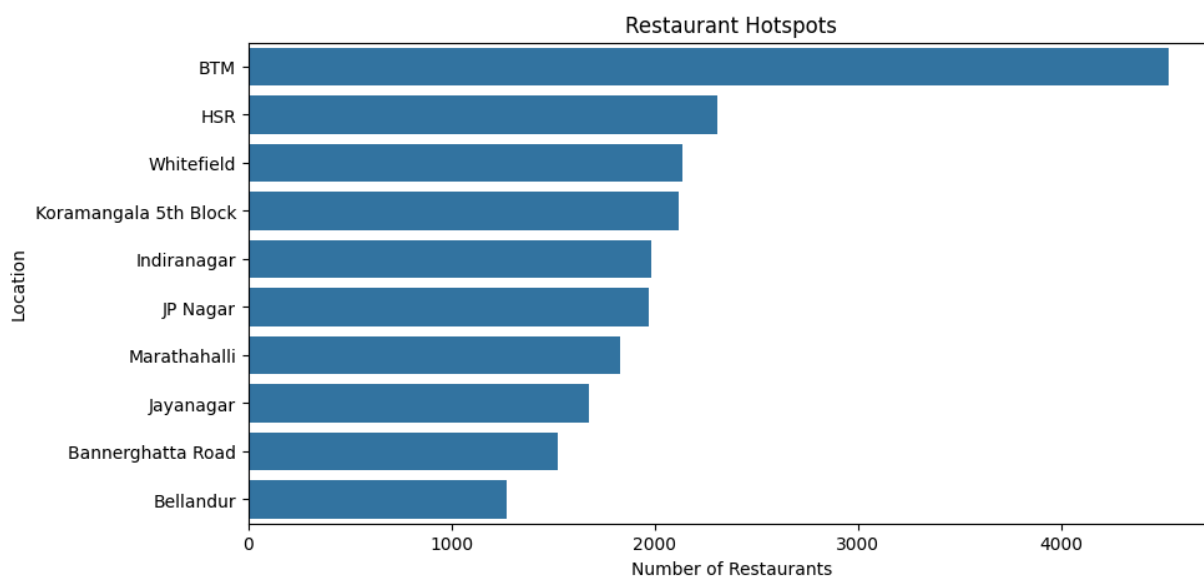


```
print(df.columns.tolist())
```

```
['address', 'name', 'online_order', 'book_table', 'rate', 'votes', 'phone', 'location', 'rest_type']
```

```
top_locations = df['location'].value_counts().head(10)

plt.figure(figsize=(10,5))
sns.barplot(x=top_locations.values, y=top_locations.index)
plt.title("Restaurant Hotspots")
plt.xlabel("Number of Restaurants")
plt.ylabel("Location")
plt.show()
```



```
print(df.columns.tolist())
```

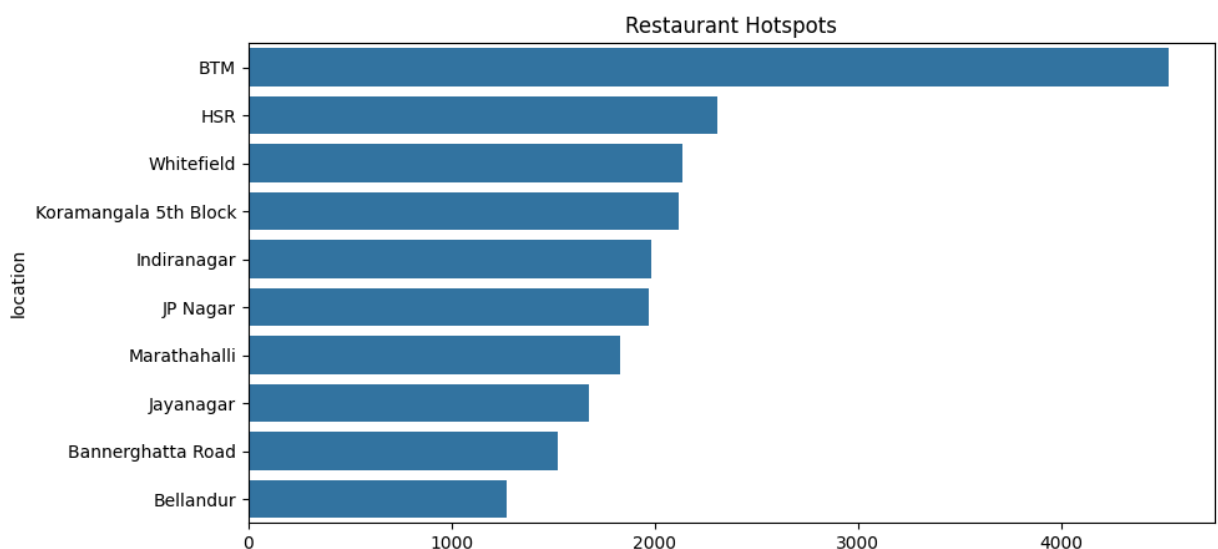
```
['address', 'name', 'online_order', 'book_table', 'rate', 'votes', 'phone', 'location', 'rest_type
```

```
for col in df.columns:
    print(col)
```

```
address
name
online_order
book_table
rate
votes
phone
location
rest_type
dish_liked
cuisines
approx_cost(for two people)
listed_in(type)
```

```
top_locations = df['location'].value_counts().head(10)

plt.figure(figsize=(10,5))
sns.barplot(x=top_locations.values, y=top_locations.index)
plt.title("Restaurant Hotspots")
plt.show()
```



```
print(df.columns.tolist())
```

```
['address', 'name', 'online_order', 'book_table', 'rate', 'votes', 'phone', 'location', 'rest_type
```

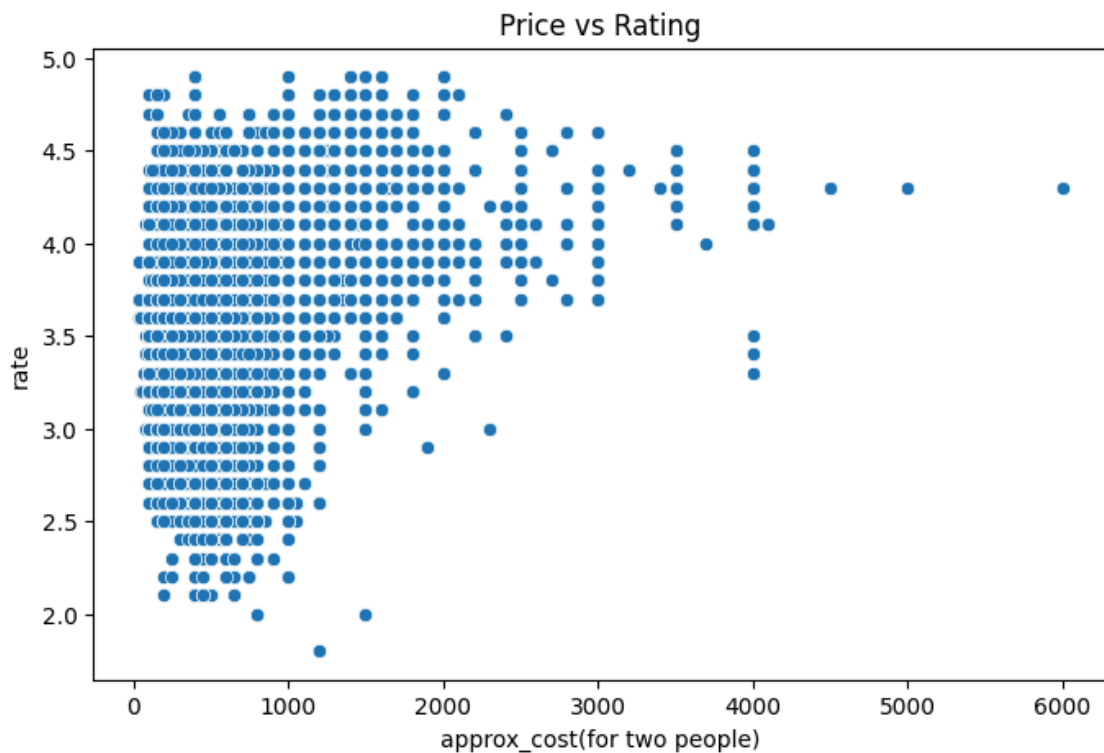
```
for col in df.columns:
    if 'cost' in col.lower() or 'price' in col.lower():
        print(col)
```

```
approx_cost(for two people)
```

```
df['approx_cost(for two people)'] = pd.to_numeric(
    df['approx_cost(for two people)'].astype(str).str.replace(',', ''),
    errors='coerce'
)

plt.figure(figsize=(8,5))
sns.scatterplot(
    x='approx_cost(for two people)',
```

```
y='rate',  
data=df  
)  
plt.title("Price vs Rating")  
plt.show()
```



```
print(df.columns.tolist())
```

```
['address', 'name', 'online_order', 'book_table', 'rate', 'votes', 'phone', 'location', 'rest_type']
```

```
plt.figure(figsize=(8,6))  
sns.heatmap(df.corr(numeric_only=True),  
            annot=True,  
            cmap='coolwarm')  
plt.title("Correlation Heatmap")  
plt.show()
```

